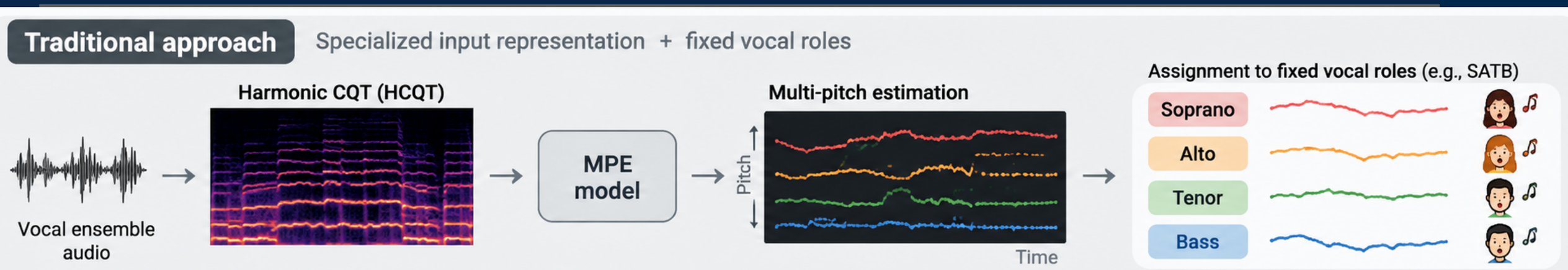
Linear STFT Multi-Pitch Estimation and Role-Agnostic Voice Attribution for Vocal Ensembles

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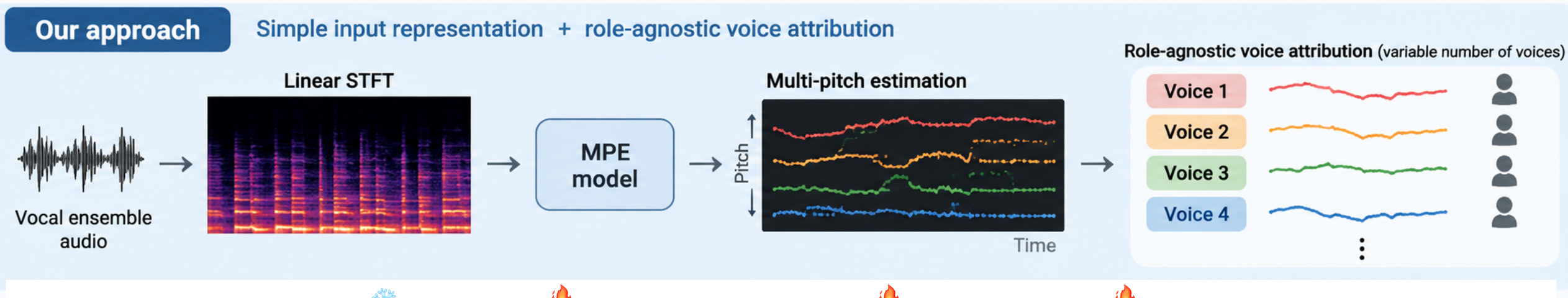
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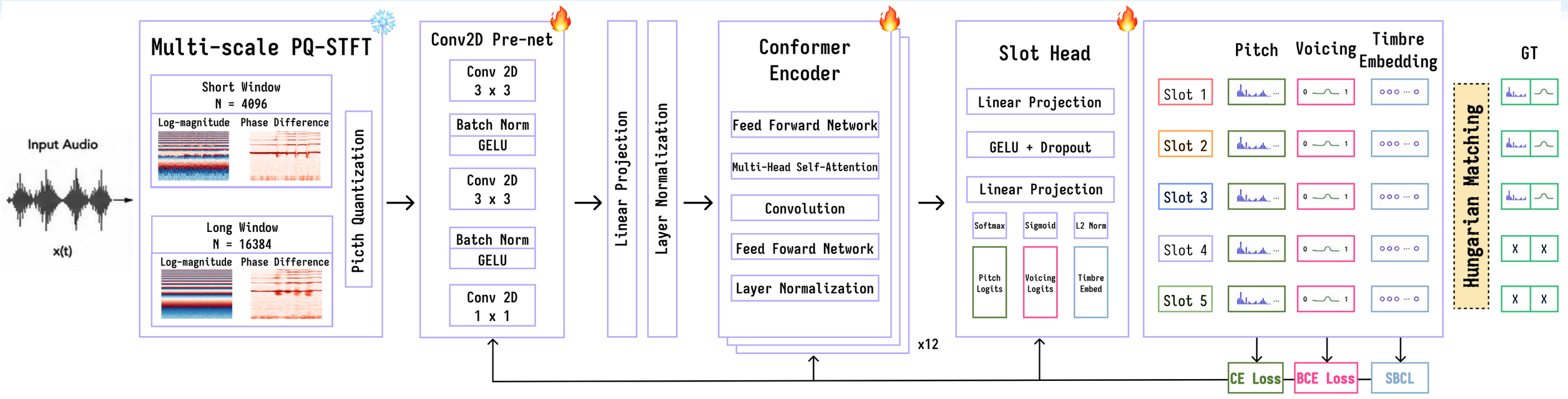
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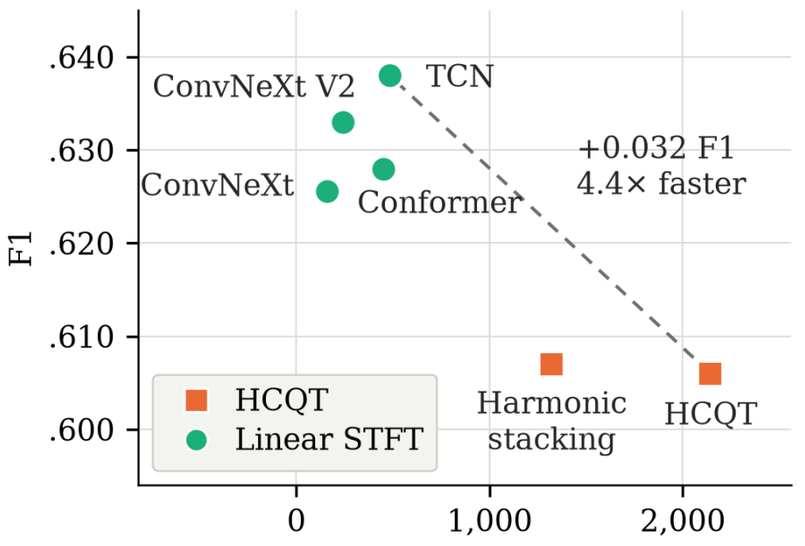
Do we need either of these assumptions?





1. Simple linear STFT outperforms HCQT.

Linear STFT consistently improves MPE accuracy across architectures, with **+0.020 pooled F1**, while substantially reducing feature-extraction cost.



2. Higher frequency resolution is not the key.

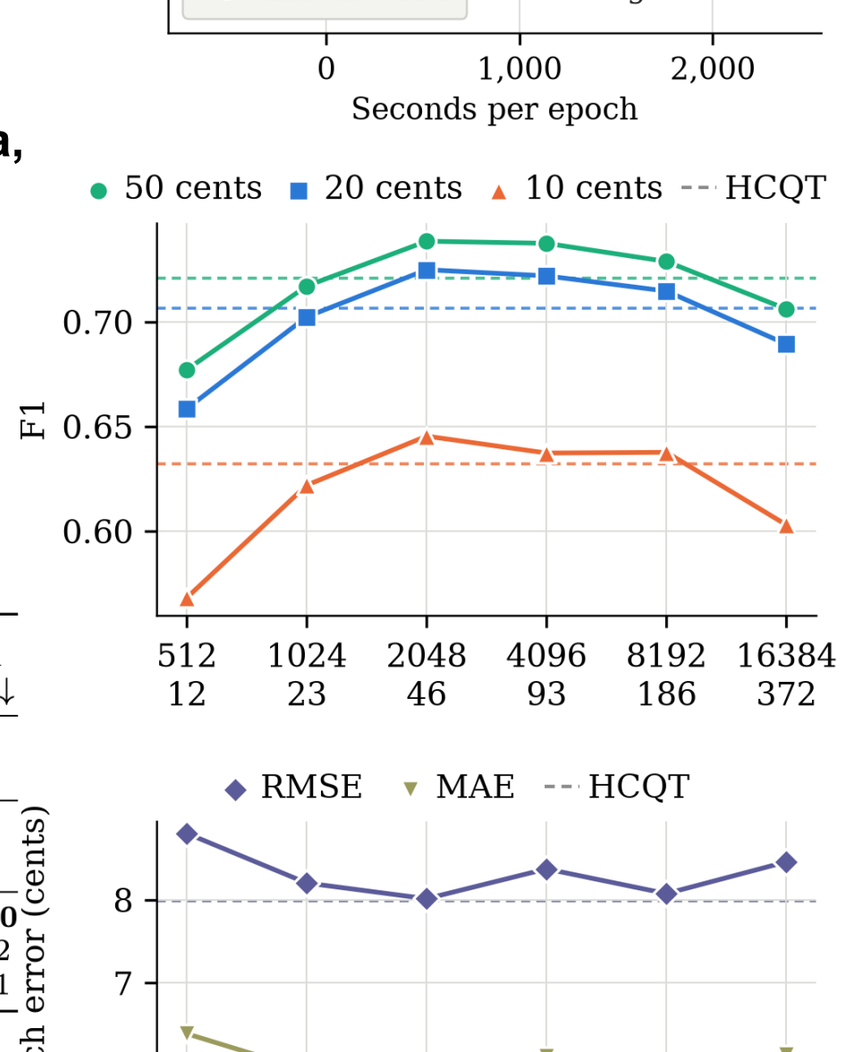
A short **46-ms STFT window performs best**; longer windows and finer frequency resolution provide no additional benefit.

3. Role-agnostic slots accurately track individual voices.

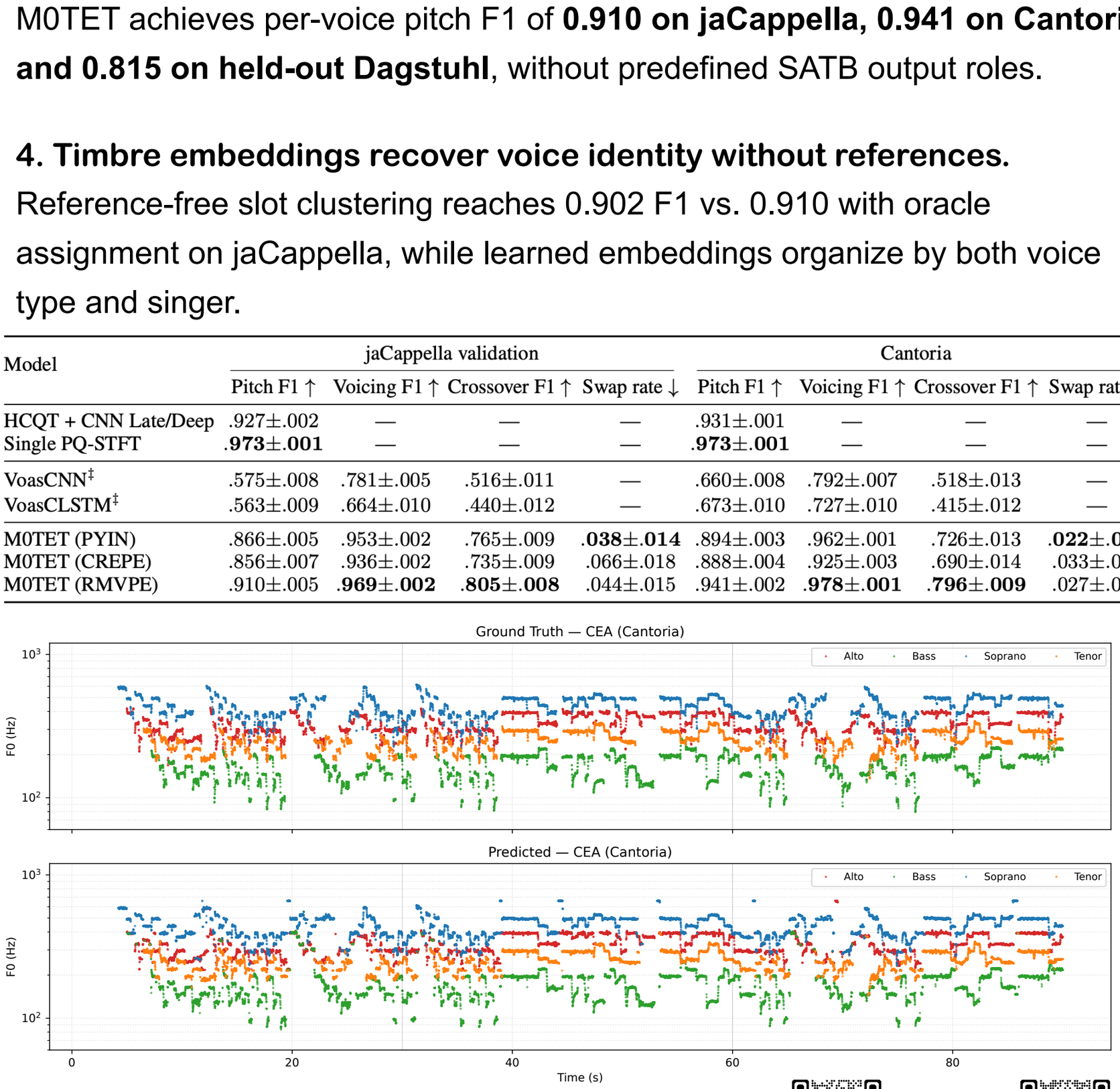
M0TET achieves per-voice pitch F1 of **0.910 on jaCappella, 0.941 on Cantoria, and 0.815 on held-out Dagstuhl**, without predefined SATB output roles.

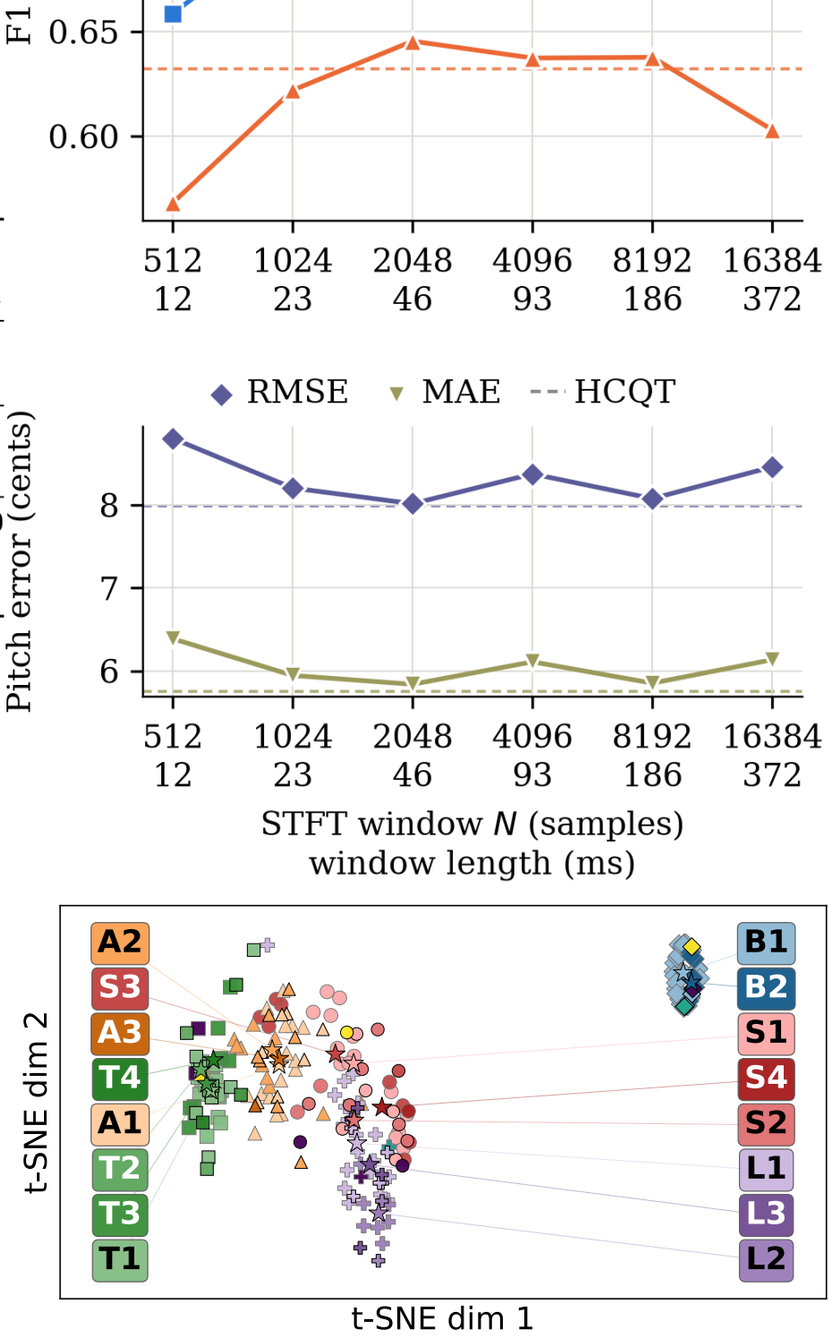
4. Timbre embeddings recover voice identity without references.

Reference-free slot clustering reaches 0.902 F1 vs. 0.910 with oracle assignment on jaCappella, while learned embeddings organize by both voice type and singer.



Model	jaCappella validation				Cantoria			
	Pitch F1 ↑	Voicing F1 ↑	Crossover F1 ↑	Swap rate ↓	Pitch F1 ↑	Voicing F1 ↑	Crossover F1 ↑	Swap rate ↓
HCQT + CNN Late/Deep	.927±.002	—	—	—	.931±.001	—	—	—
Single PQ-STFT	**.973±.001**	—	—	—	**.973±.001**	—	—	—
VoasCNN‡	.575±.008	.781±.005	.516±.011	—	.660±.008	.792±.007	.518±.013	—
VoasCLSTM‡	.563±.009	.664±.010	.440±.012	—	.673±.010	.727±.010	.415±.012	—
M0TET (PYIN)	.866±.005	.953±.002	.765±.009	**.038±.014**	.894±.003	.962±.001	.726±.013	**.022±.010**
M0TET (CREPE)	.856±.007	.936±.002	.735±.009	.066±.018	.888±.004	.925±.003	.690±.014	.033±.012
M0TET (RMVPE)	.910±.005	**.969±.002**	**.805±.008**	.044±.015	.941±.002	**.978±.001**	**.796±.009**	.027±.011





(b) Singer: color = singer within a part
within-part retrieval 70% vs 58% majority null

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